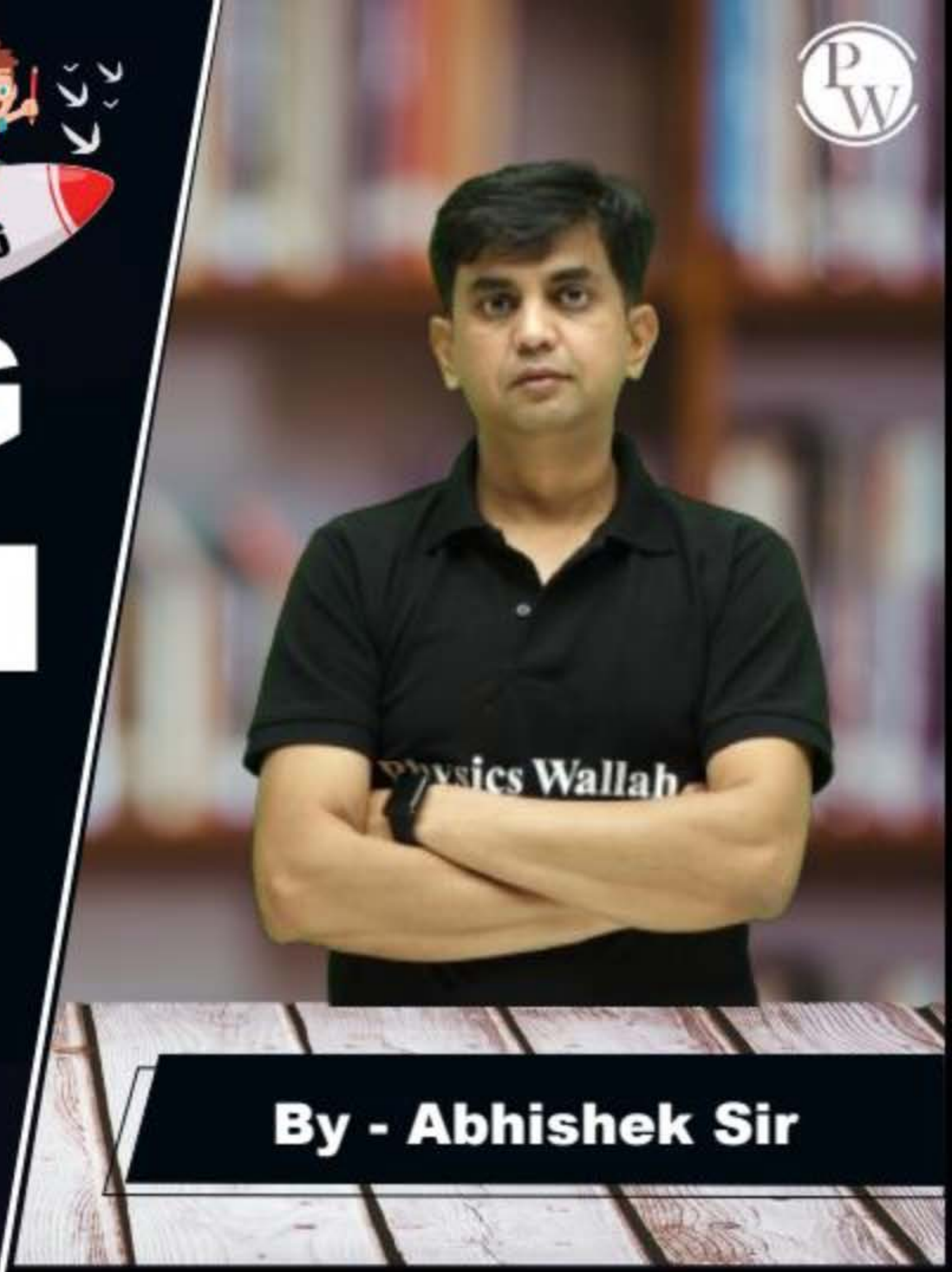
CS & IT ENGINEERING

Computer Network

Introduction

Lecture No. - 04



By - Abhishek Sir

Recap of Previous Lecture

Topic Transport Layer

Topics to be Covered

Topic Transport Layer

Topic Network Layer

Topic Data Link Layer

ABOUT ME

Hello, I'm Abhishek

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

Telegram Link : https://t.me/abhisheksirCS_PW

@ABHISHEKSIRCS_PW

#Q. Assume that source S and destination D are connected through two intermediate routers labeled R. Determine how many times each packet has to visit the network layer and the data link layer during a transmission from S to D ?

[GATE-2013, 1-Mark]

IIT-B

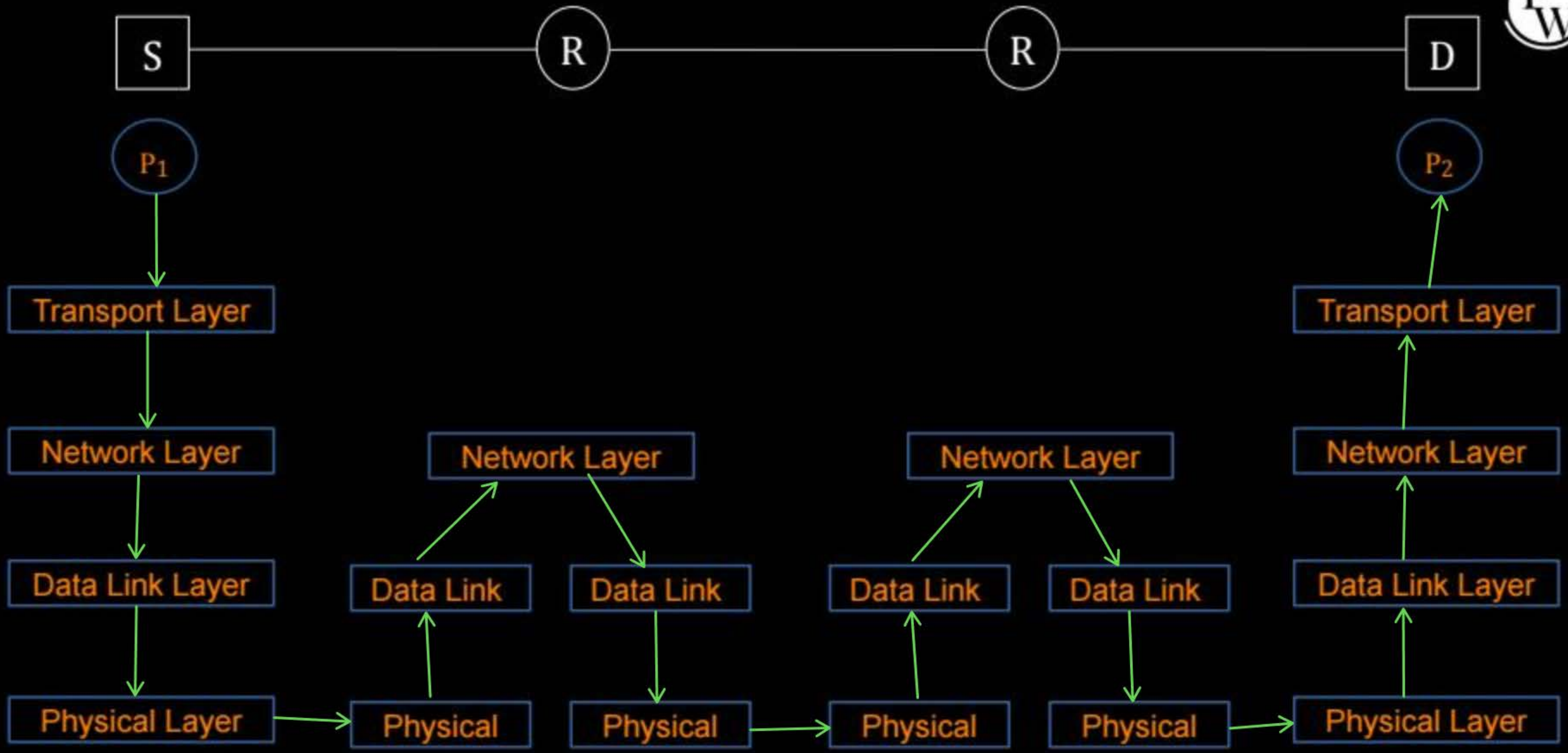
(A) Network layer – 4 times and Data link layer – 4 times

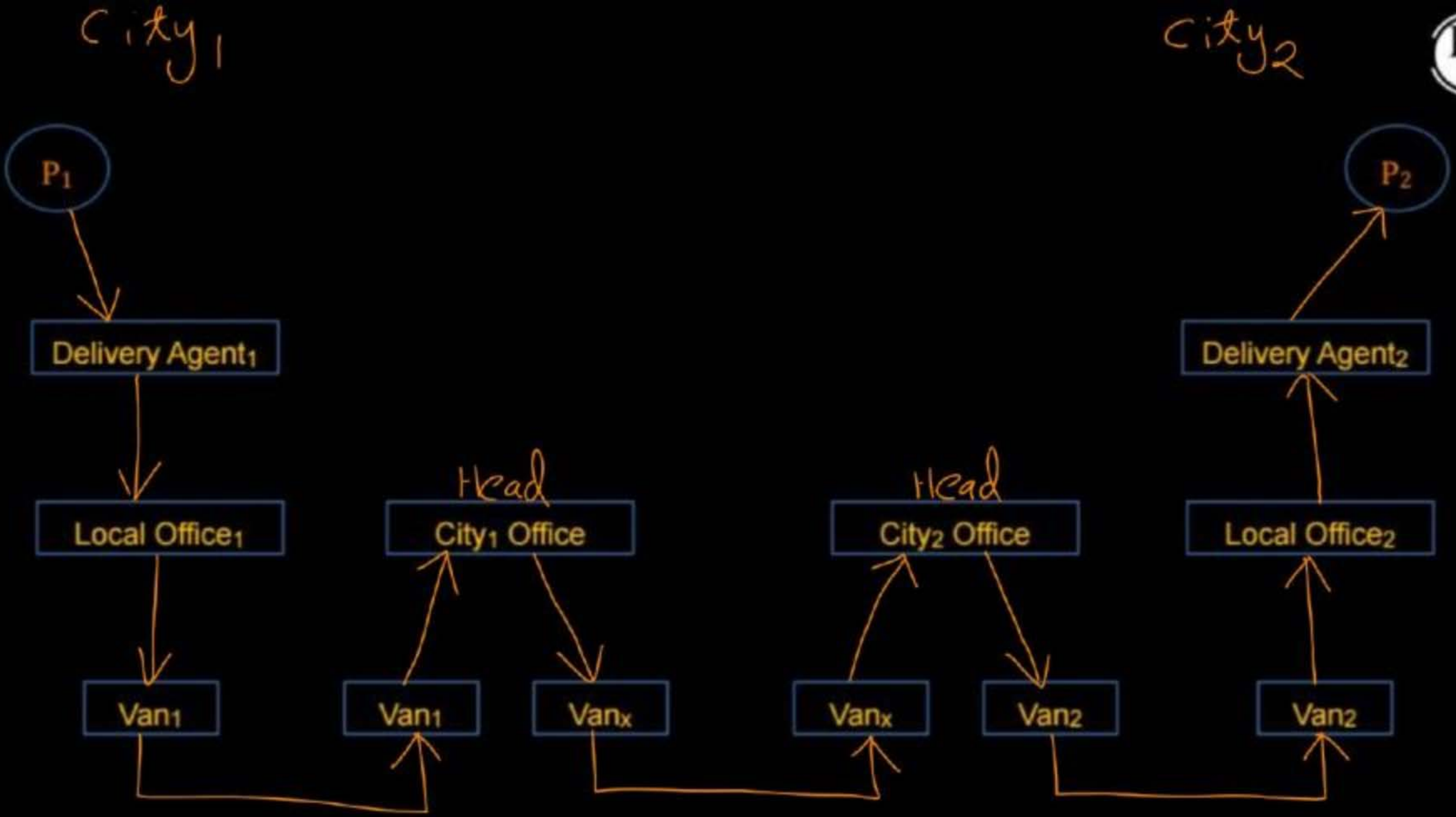
(B) Network layer – 4 times and Data link layer – 3 times

(C) Network layer – 4 times and Data link layer – 6 times

(D) Network layer – 2 times and Data link layer – 6 times

$\boxed{\text{Ans: C}}$


S
R
R
D
P1
P2
Transport Layer
Transport Layer
Network Layer
Network Layer
Network Layer
Network Layer
Data Link Layer
Data Link
Data Link
Data Link
Data Link
Data Link Layer
Physical Layer
Physical
Physical
Physical
Physical
Physical Layer



City1
City2
P1
P2
Delivery Agent1
Delivery Agent2
Local Office1
Head
City1 Office
Head
City2 Office
Local Office2
Van1
Van1
Vanx
Vanx
Van2
Van2

Topic : Transport Layer PDU *

-> In 'Internet Protocol Suite' (TCP/IP Model)

1. For TCP : "Segment"
2. For UDP : "Datagram"

-> In 'OSI Model'

Transport Layer PDU : "Segment" ✓

[For both TCP and UDP]

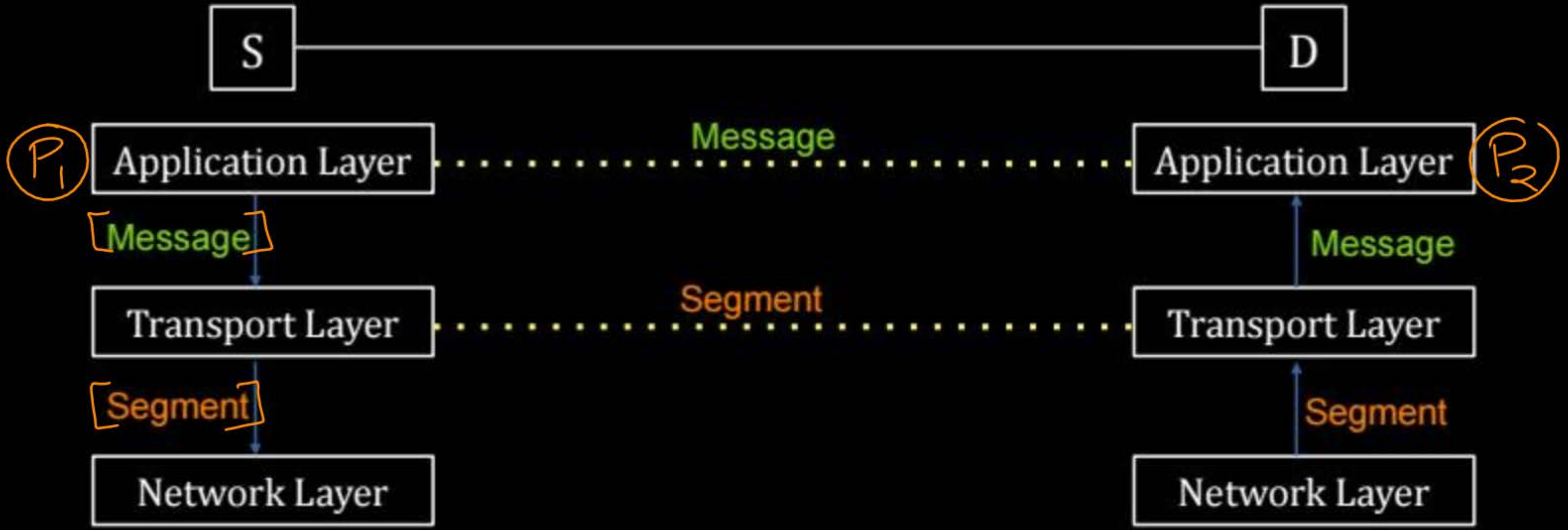
Topic : Transport Layer

-> Transport Layer PDU : "Segment"

-> Sender : Divide application messages into segments, Segments passes to network layer → $\boxed{\text{Segmentation}}$

-> Receiver : Resembles segments into messages, Messages passes to application layer → $\boxed{\text{Resembly}}$

Topic : Protocol Data Unit



Topic : SDU

→ Service Data Unit (SDU)

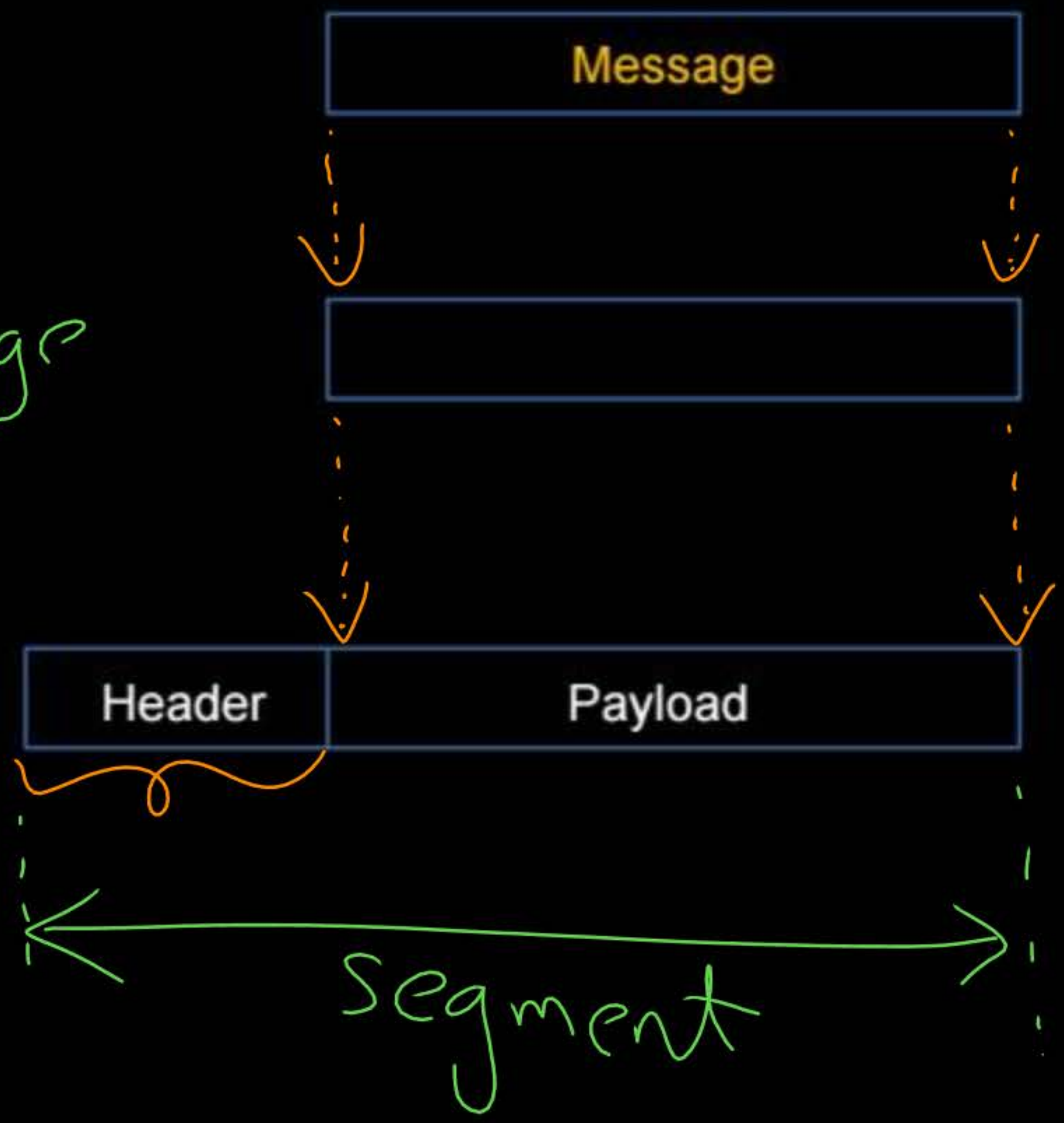
→ Upper layer 'Protocol Data Unit' (PDU)

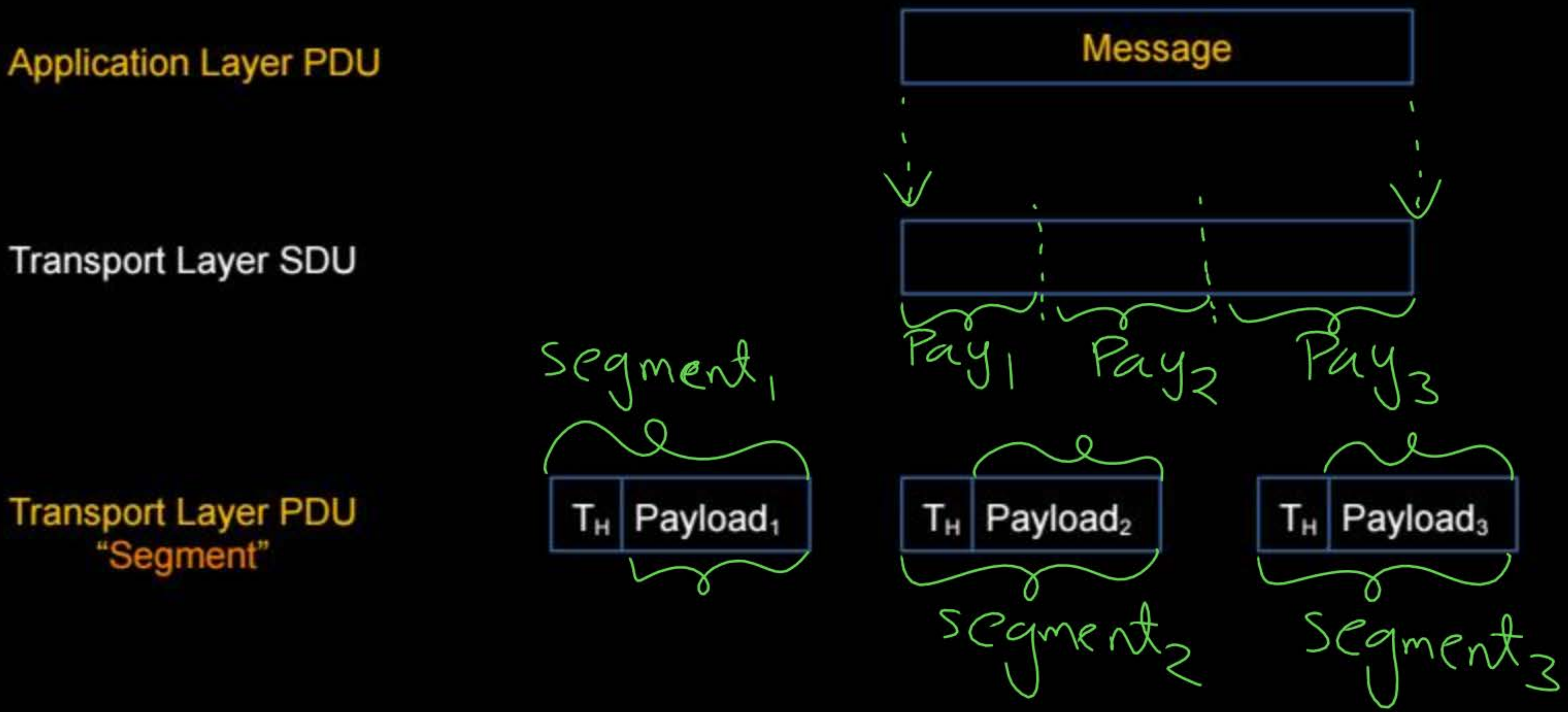
→ Layer n PDU is SDU for Layer (n–1)

Application Layer PDU

Transport Layer SDU = Message

Transport Layer PDU
"Segment"





Application Layer PDU
Message
Transport Layer SDU
Pay1
Pay2
Pay3
Transport Layer PDU
"Segment"
segment1
TH
Payload1
TH
Payload2
TH
Payload3
segment2
segment3

#Q. What is the maximum size of data that the application layer can pass on to the TCP layer below?

[GATE-2008]

IISC

(A) Any Size

(B) 2^{16} bytes - size of TCP Header

(C) 2^{16} bytes

(D) 1500 bytes

$\boxed{\text{Ans: A}}$

#Q. What is the maximum size of data that the application layer can pass on to the TCP layer below?

[GATE-2008]

(A) Any Size

(B) 2^{16} bytes - size of TCP Header

(C) 2^{16} bytes

(D) 1500 bytes

Ans : (A) Any Size

Application layer can send any size of data.

No any limit defined by OSI Model.

Topic : Network Layer

-> Provide host-to-host connectivity

-> Forwarding and Routing

-> Internet protocol (IP)

1) IPv4

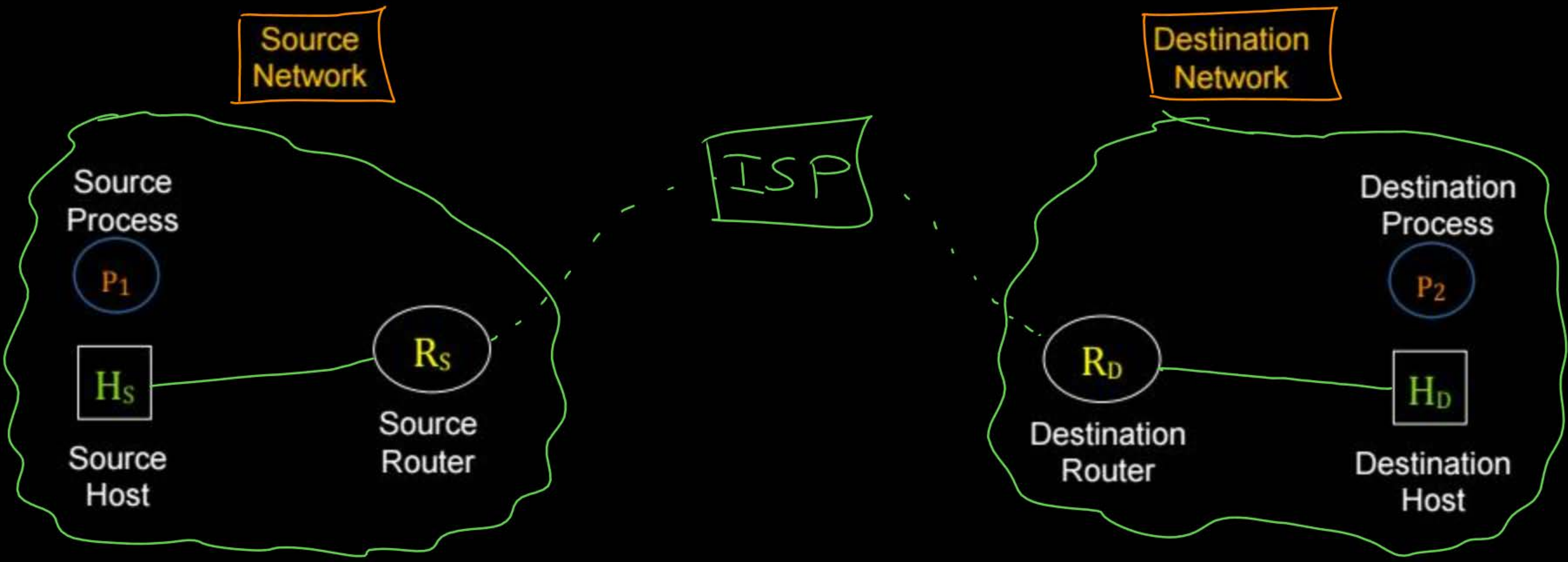
2) IPv6

Difference b/w internet and Internet

H.W.

Topic : Host-to-Host Connectivity

inter-networks : Source & Destination hosts belongs to different networks



Topic : Host-to-Host Connectivity

→ Routing of IP packets from source host to destination host.

1. Source host to source router
2. Source router to destination router ⟹ [Routing]
3. Destination router to destination host

Topic : Forwarding

Data Plane :

-> Determine how IP packets are forwarded

[Using Router's Forwarding/Routing table]

-> Move packet from a router's input link to appropriate router's output link

Topic : Routing

Control Plane :

-> Determine how IP packets routed among routers

[Determine route taken by IP packets from source to destination]

-> Construction of Router's Forwarding/Routing table

[Using Routing algorithms/protocols]

Topic : Network Layer PDU

-> In 'Internet Protocol Suite' (TCP/IP Model) ⇒ Packet-switched Network

Network Layer PDU : "Packet" ✓
(Internet)

-> In 'OSI Model'

Network Layer PDU : $\boxed{\text{"Datagram"}}$ ✓ or "Packet"

IP Datagram / IP Packet / IP Fragment

Topic : Network Layer

[IP Datagram]

Network Layer PDU : "**Datagram**"

-> Sender : Divide segments into datagrams ,
Datagrams passes to data link layer

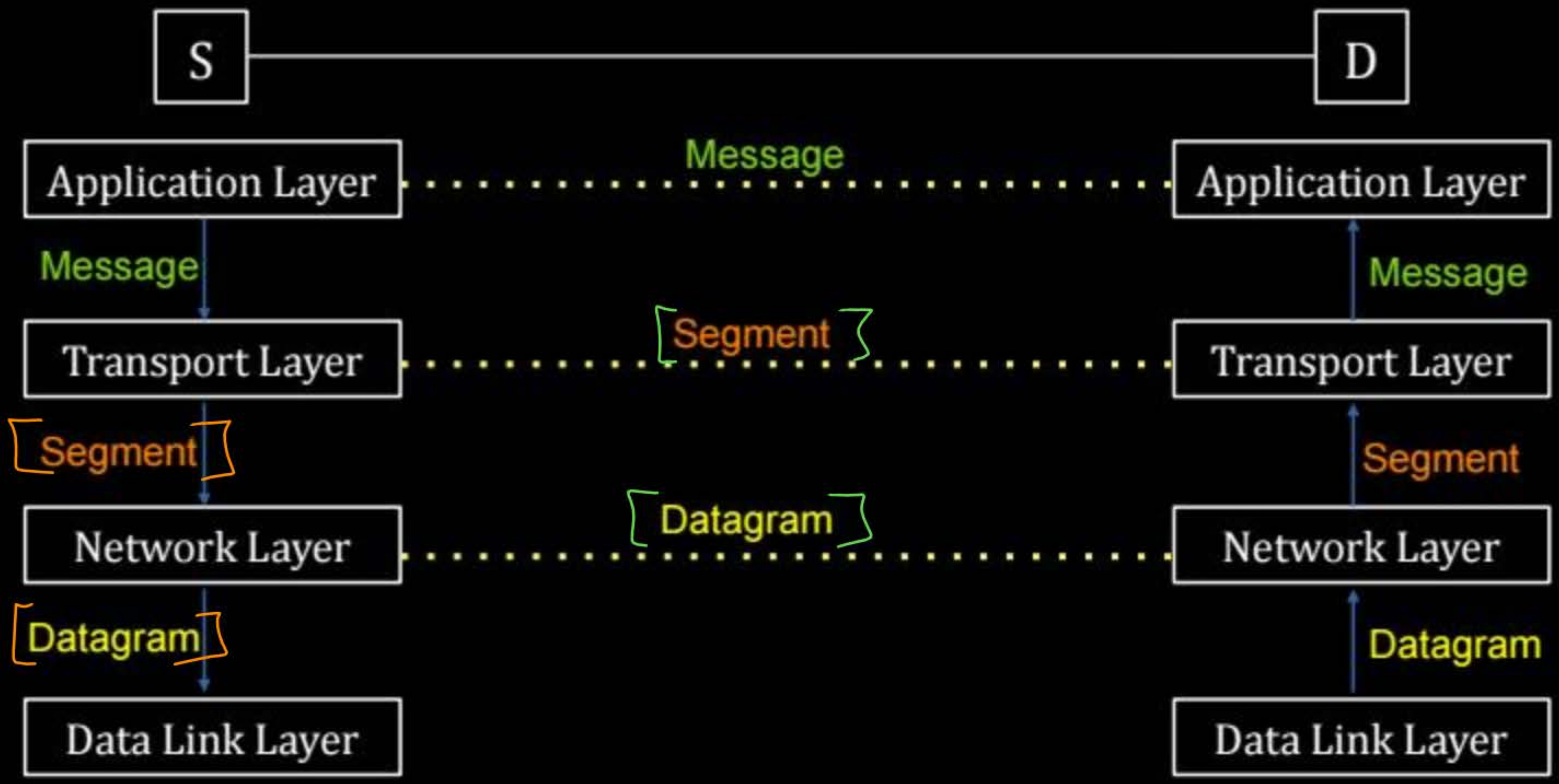
$\}$ → $\boxed{\text{Fragmentation}}$

-> Receiver : Resembles datagrams into segments,
Segments passes to transport layer

$\}$ → $\boxed{\text{Resembly}}$

[Reassembly]

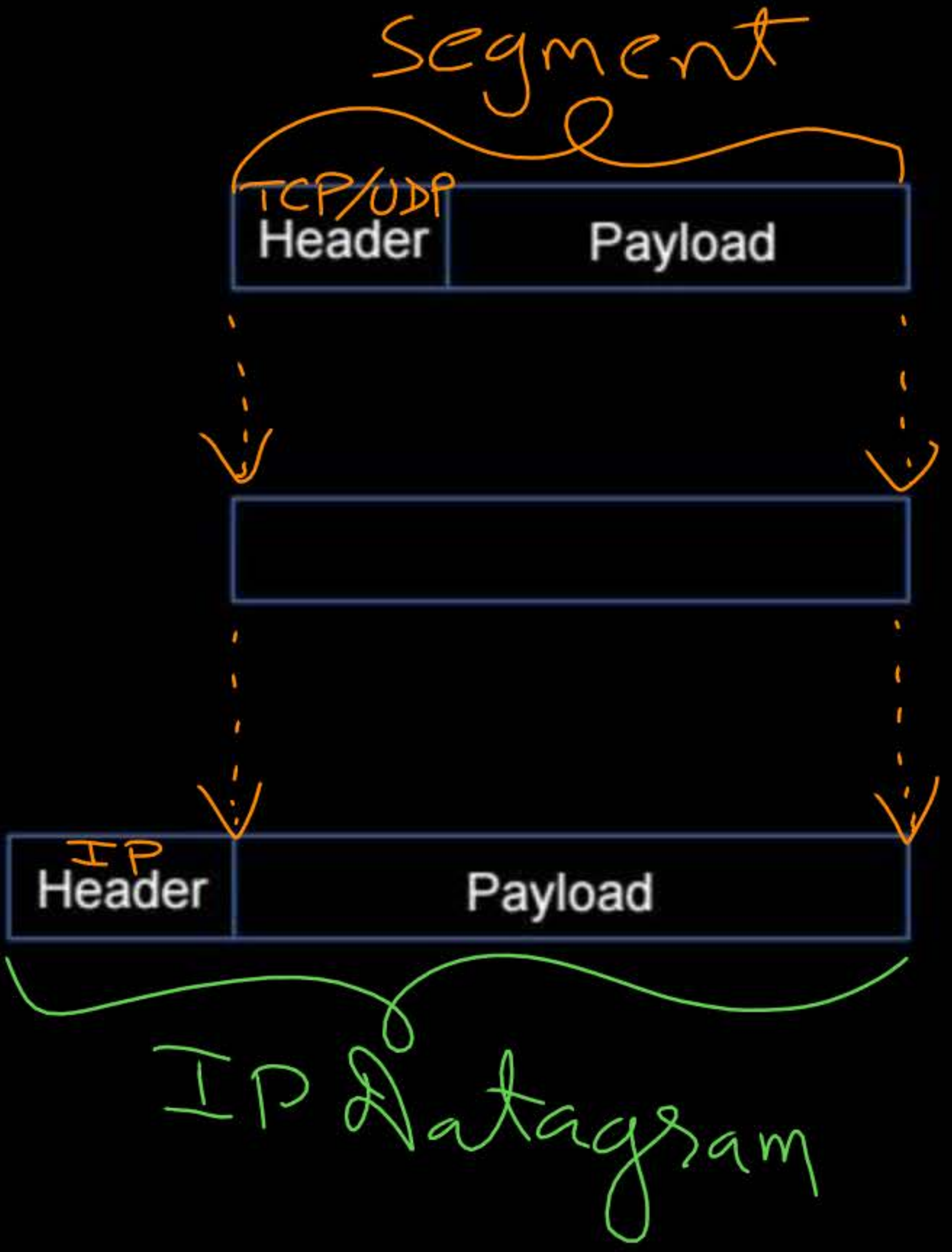
Topic : Protocol Data Unit



Transport Layer PDU
"Segment"

Network Layer SDU = "segment"

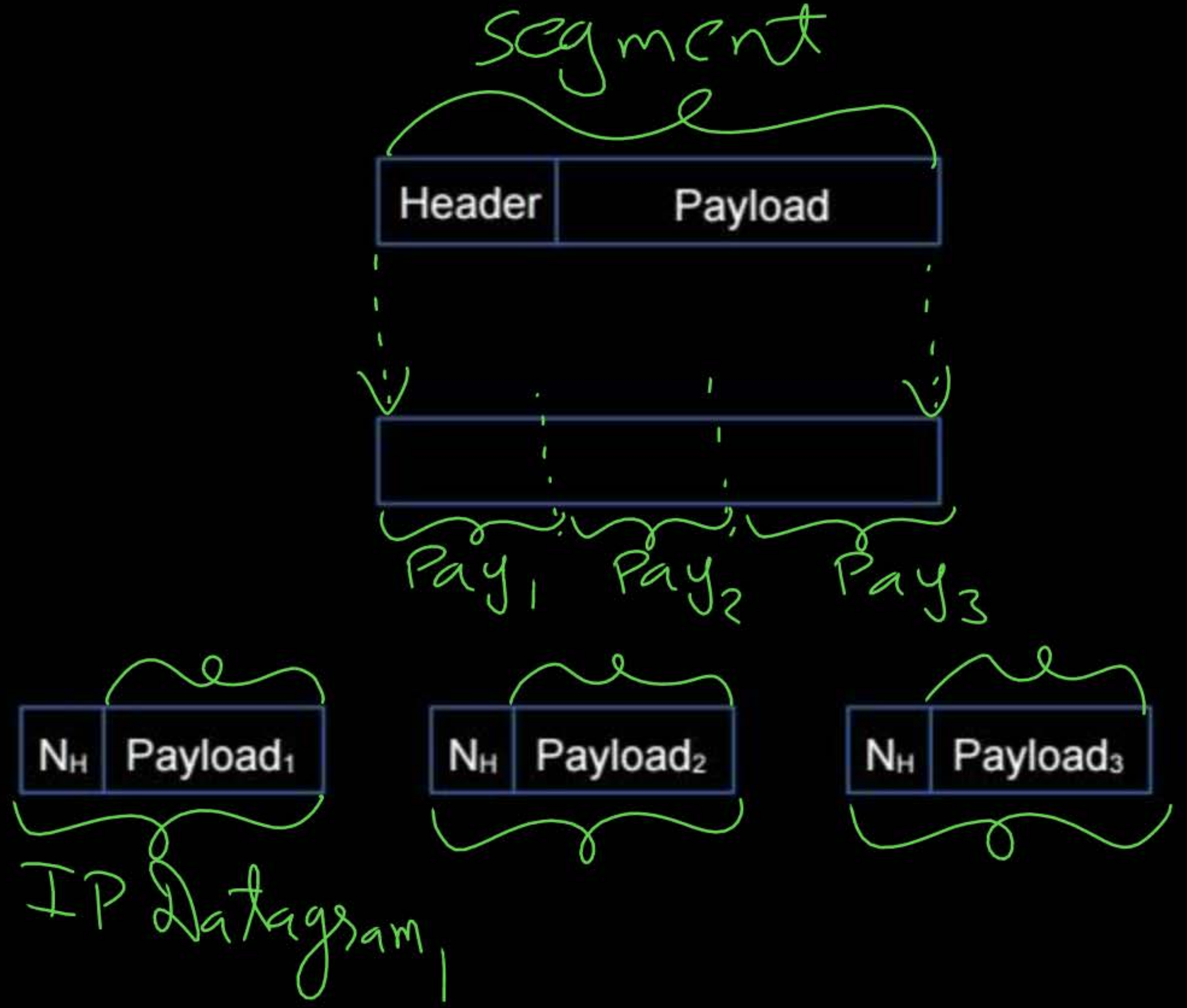
Network Layer PDU
"Datagram"



Transport Layer PDU
"Segment"

Network Layer SDU

Network Layer PDU
"Datagram"



Topic : Network Layer

-> Network Layer Networking Device : $\boxed{\text{"Router"}}$ ✓ [Layer-3 device]

-> Store and Forward device

[Store, Process and Forward]

-> Forwarding based on IP Address

→ No any Networking Device for transport Layer

→ Gateway: Application Layer (Layer-7) networking device.

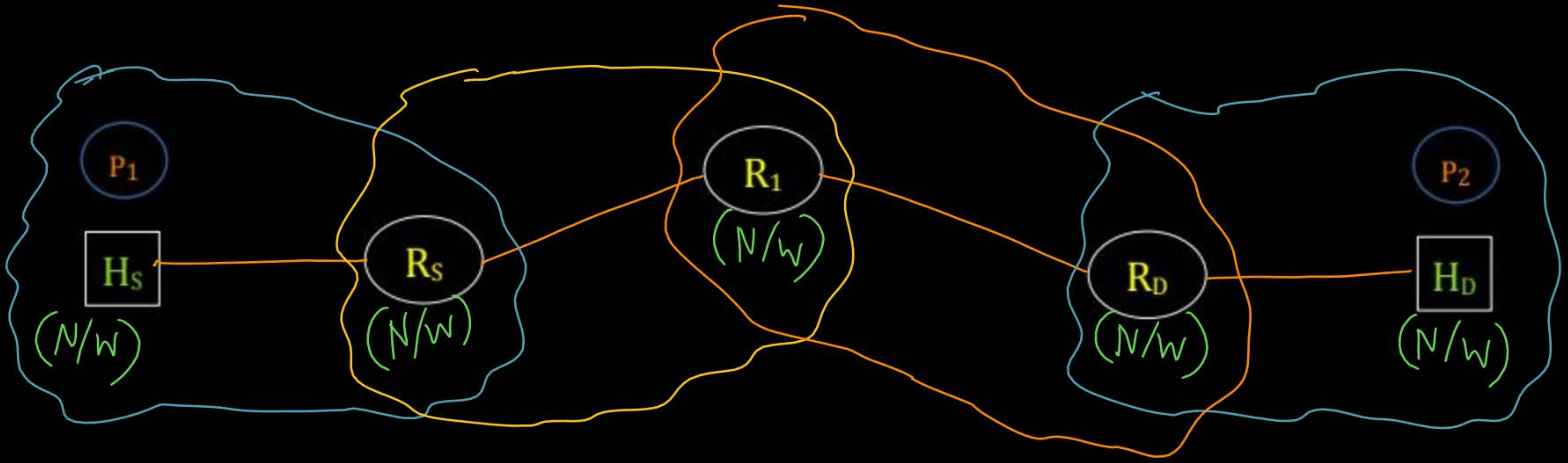
Topic : Data Link Layer

-> Provide node-to-node connectivity

Topic : Node-to-Node Connectivity

Intra-network : Source & Destination hosts belongs to same network [within a network]

Node : Host or Router (Any Layer-3 device and above)



2 mins Summary

Topic — Transport Layer

Topic — Network Layer

Topic — Data Link Layer

THANK - YOU

@ABHISHEKSIRCS_PW